# PREM M

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## PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with a 9.12 CGPA and skills in SQL, Python, Java, C, DSA, and IoT. Hands-on experience with Arduino, ESP32, Firebase, and real-time data monitoring. Seeking a Cloud Data Engineer Trainee role to apply my technical and problem-solving skills while building expertise in data engineering.

## EDUCATION

**B.E. – Electronics and Communication Engineering** | Nandha Engineering College, Erode | 2023– Present | CGPA: 9.12
**HSC (12th)** | Kamaraj Municipal Boys Higher Secondary School, Erode | 2022–2023 | 94%

## TECHNICAL SKILLS

**Programming:** Java, Python (Basics), C (Basics)
**Data & Databases:** SQL, Data Structures and Algorithms

## PROJECTS

**Smart Pharmacy Queue and Service Management System**
- **IoT | ESP32 | Firebase | Embedded C/C++ | HTML/CSS/JavaScript**
- Developed an IoT-based pharmacy queue management system using ESP32 to organize customer service and reduce waiting time.
- Implemented priority-based queue handling using a keypad, LCD display, control buttons, buzzer, and LED indicators.
- Integrated Firebase for real-time storage and monitoring of customer, prescription, and queue information through a web dashboard.
- Added prescription refill reminders and real-time queue status updates to support efficient pharmacy service management.

**Smart Hairpin Bend Accident Prevention System**
- Designed a system to reduce accidents at hairpin bends on hill roads.
- Used sensors to detect vehicles approaching from both sides of the bend.
- Implemented warning signals using LED and buzzer alerts to help prevent collisions.

**Vehicle Theft Detection Using Vibrational Sensor**
- Developed a security system to detect unauthorized movement or tampering of vehicles.
- Used a vibration sensor to detect unusual motion.
- Triggered an alarm/alert when vibration exceeded a defined threshold.

## INTERNSHIP EXPERIENCE

**Embedded Programming on Arduino AVR – Beginner | Embuzz Technologies** | 16 June 2025 – 30 June 2025
- Worked on embedded systems and microcontrollers, including Arduino UNO programming and sensor interfacing.
- Developed mini projects using LEDs, buzzers, and sensors.

**Embedded Systems & IoT Internship | Caliber Embedded Technologies** | 23 December 2025 – 05 January 2026
- Worked on IoT-enabled sensor monitoring systems.
- Integrated Bluetooth HC-05 modules and multiple sensors.
- Tested and analyzed real-time hardware data for embedded applications.

## CERTIFICATIONS & ACHIEVEMENTS

- Cloud Computing – NPTEL (76%)
- Internet of Things (IoT) – NPTEL (76%)
- Ethical Hacking – NPTEL (62%)
- Honours Diploma in Computer Applications – TCEDS (96%)
- Technical Quiz – Vidyaa Vikas College of Engineering and Technology (III Prize)

## ADDITIONAL INFORMATION

- Core strengths: Problem Solving, Learnability, Team Collaboration, Communication, Attention to Detail
- Career focus: Cloud Data Engineering, SQL, Python, Data Processing, Cloud and IoT Systems